CHB x BCA Case Competition 2025

Pharmora P.U.L.S.E.

J - LAMB

Presented by: Bruno Nguyen, Andy Dang, Janice Lin, Leyna Nguyen, Avelyn Nguyen

Pumping and Circulating Pharmora's Drugs Onto Customers

Executive Summary

Situation

1 **Struggled to match the pace of competitors** and adapt to the fast-changing of pharma industry regarding new regulations, technologies, and data-driven approach

2 **Narrow demographic focus** and go-to-market (GTM) **strategy are underdeveloped**

3 Experiencing a **2.33% decline in YoY revenue growth** in 2023-2024 compared to 2022-2023

Question

"How can Pharmora recover from a financial downturn and long-term, retrieve its competitiveness in the pharmaceutical market?"

P.U.L.S.E. Strategy

- **P**osition in Targeted Markets
- **U**tilize Distribution Channels
- **L**aunch Promotional Campaigns
- **S**erving Patients With Convenience
- **E**nforce AI into Drug Development Processes

Impact

Up to 47%

Increase in net revenue

2-3%

OPEX reduction

More than 50%

Increase total obtainable customers

Commercial Challenge: Pharmora relies on a single partner for a critical commercial function, limiting its customer reach

Company Background & Problem Analysis

- 45 years
- Cardiovascular franchise + diabetes & obesity

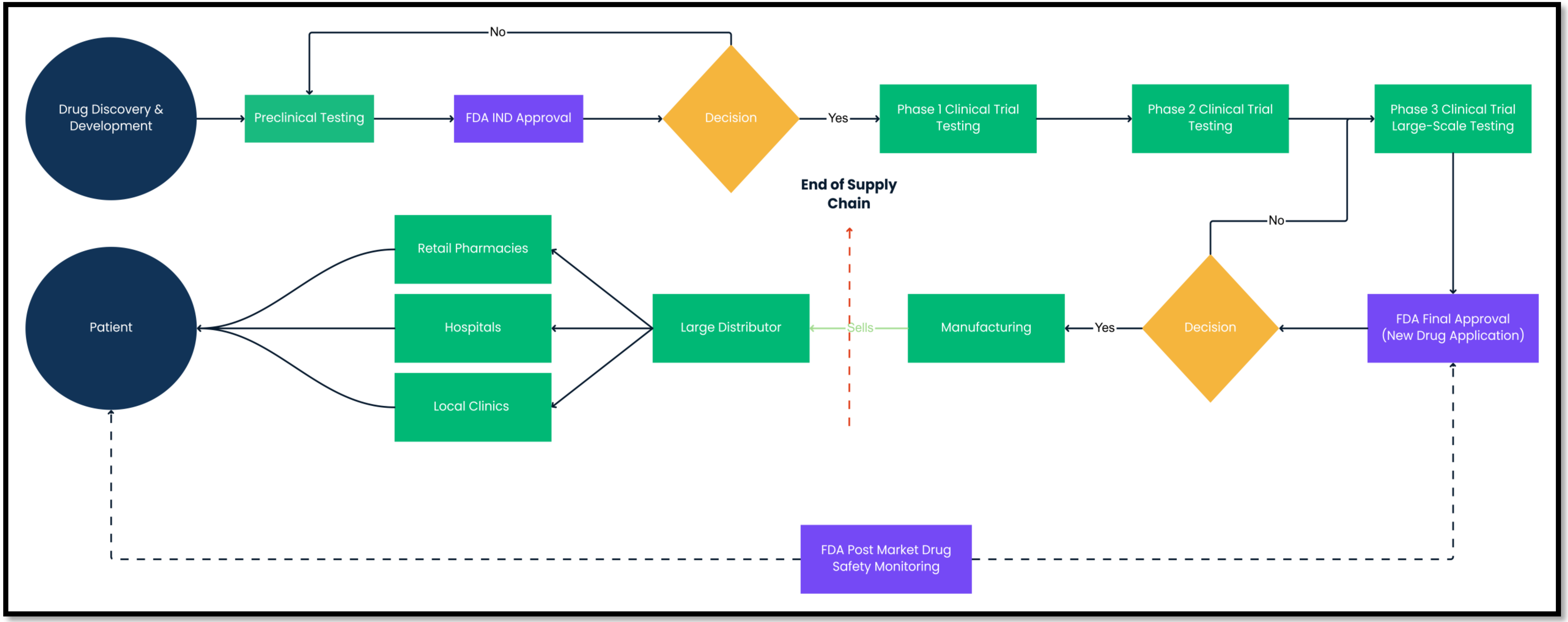
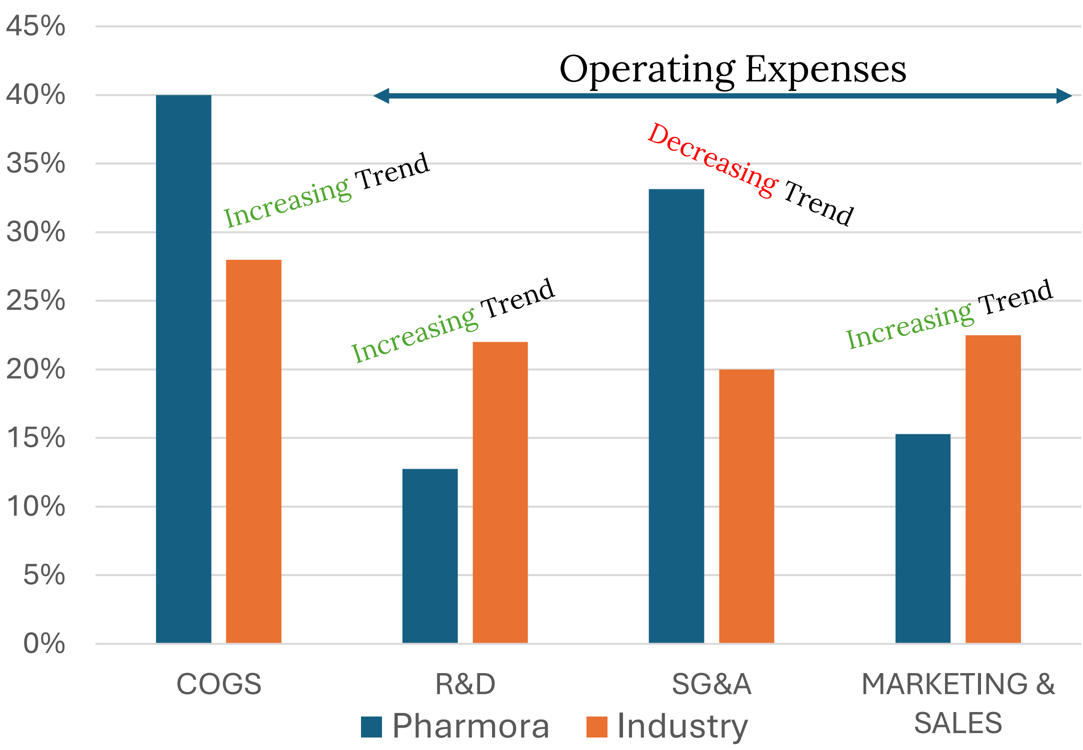


Figure 1: ***Flow Chart of Pharmora's Supply Chain (FDA.gov)***

Operational Inefficiency: Pharmora should reallocate its expenses

Company Background & Problem Analysis

Pharmora vs. Pharma Industry Expense Breakdown (% Total Revenue)



Cost Of Goods Sold (COGS)

Gap: Higher manufacturing costs vs. industry benchmark
Impact: Lower margins and reduced competitiveness

Research & Development (R&D)

Gap: Underinvestment in innovation pipeline
Impact: Weaker long-term product development capacity

Selling, General & Administrative (SG&A)

Gap: SG&A significantly exceeds industry norms when adding marketing
Impact: Indicates heavy overhead and inefficient cost structure

Marketing & Sales

Gap: Marketing spending is below peers
Impact: Insufficient provider education and slower revenue growth

Implications:

To stay competitive, lower SG&A, spend more R&D and allocate the expenses efficiently, and lower COGS to a minimum.

Source: [1] Maven Bio research (2024), [2] Bird Marketing (2025)

Pharmora should bridge the innovation gap to its competitors.

Company Background & Problem Analysis

	Ash & Co.	Cardiva	BioNova	Firenza
Research & Development	Generative AI		R&D Cost-Sharing	
Manufacture	Automation Systems & U.S.-Based Manufacturing Hub			
Launch & Market		Partner with Private Hospitals	Expand Primary Care & Retail Channels	
Products			License Drug Candidates Overseas	Affordable Oral Obesity Formulations
Chronic Care		AI Clinical Design & Remote Monitoring Platforms		Digital Engagement Platforms
Deployment Speed	Moderate-Slow	Moderate-Fast	Moderate-Fast	Fast
Impact to Pharmora	Moderate-Low	Moderate-High	Moderate	High

Implications: Pharmora is facing threats from its competitors in the next five years due to customer base expansion, partnerships, and technological advancement

Pharmora Is In A Crucial Transitional Phase

Market Analysis – Competitive Landscape

Pharmora

Operational Inefficiency

Innovation Gap

Commercialization Challenge

Government

Increase adoption in Health Technology Assessments (HTA)
Transparency becomes increasingly important

Inflation Reduction Act (IRA) & Most-Favoured-Nation (MFN) policies
Increase downward pressure on pharmaceutical companies through a price cap and price negotiation

Suppliers

Possibility of a 100% tariff on overseas developed drugs unless manufactured within the US
Likely increase COGS

Technological Innovation (Cigna Healthcare, 2025)

AI plays an important role in both manufacturing and servicing – Healthcare leaders

30%
increase in diagnostic accuracy due to AI-driven models

82%
have/plan to implement AI governance structures

92%
say AI will improve efficiencies

65%
say AI enables faster decision-making

P.U.L.S.E. Is A Dynamic Solution That Defibrillate Pharmora's Financial And Regain The Company's Competitiveness

Solution Framework

H O W

Should Pharmora Narrow Its Advertisement To Target Desirable Markets?	→	**P**osition in targeted markets
Can Pharmora Directly Sell Its Drugs To End Customers?	→	**U**tilize distribution channels
Can Pharmora leverage social media algorithms and create an active health community between HCPs and patients?	→	**L**aunch community campaigns
Can Pharmora Improve Its Customer Care And Provide Services Catered To Customer Demand?	→	**S**erve Patients
Can Pharmora Increase Its Supply Chain Productivity, 40% COGS, And OPEX?	→	**E**nforce AI

Pharmora

Positioning in Targeted Markets

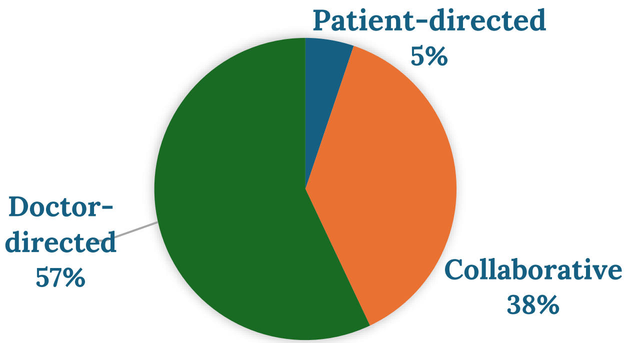
Pharmora's Opportunity Lies In Approaching New Target Markets

Market Analysis – Target Demographic

Pharmora should expand outreach to both Healthcare Professionals and patients, prioritizing Healthcare Professionals since they drive prescribing decisions.

Healthcare Professionals (HCPs)

- Educate others and patients about treatments—proper use and side effects—which requires a guideline conference
- Most trusted and influential resource in a patient's care path, their recommendation plays a central role in the decision to purchase and use a medication



Patients

- Trust their doctor's expertise to buy drugs, especially prescription drugs; the doctor's recommendation is a primary driver
- Patient Involvement Trend: **the concept of "shared decision-making' is growing, more patients want to be involved in the process**

HCPs		Patients
94%	Use smartphones for professional purposes	80%
80%	Highly value personalized content	85%
81%	Evaluate a drug's value by the pharma's reputation	58%
68%	Feel "burnt out" by pharma emails	55%

Pharmora's Market Potential Is Maximized By Customers' Trust With New Approaches

Solution (1/4) - A: Applying Digital Marketing in Normal Operation Model

1. **Normal Operation Model**
2. Direct-to-customer (DTC) Model

Digital

Onsite

Launching Product Model
Live Low. Live Strong

How can we build trust with the HCPs and Patients?

High-valued and digestible INFORMATION

Personalized Communication

Gen AI: address people by **name, key points from recent interactions, specialty, the specific HCP's need**

Contextual Communication: 3R

Right Message, Right Time, Right Channel:

- Micro-learning moment in bite-sized content (short video, infographic) (1 tip/day for a week) with the mobile-first delivery
- Omnichannel Engagement

PARTNERSHIP RELATIONSHIP

Dialogue

Social Media Platforms: LinkedIn & Twitter
Clinical Trial Insights, Join Hashtag, Engage with experts' conversations

Useful information to help HCPs care for patients

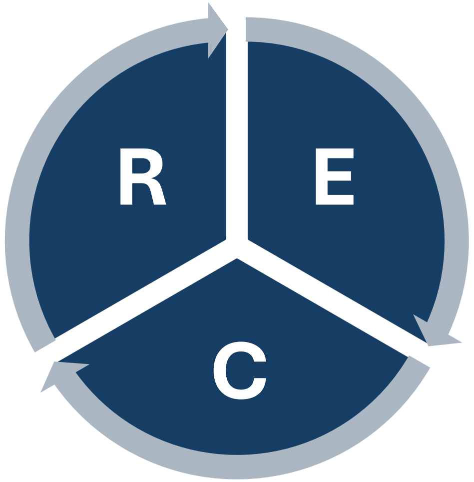
-> Great perception of value directly influences prescribing behavior
-> A relationship of value: More **willing to listen to product's message** because of the credibility as a partner in their development

1. Receive information

Make Pharmora a useful partner in helping HCPs care for patients and Patients to learn

2. Engage

Show that Pharmora listens, absorbs, and improves through connectio ns & conversations



3. Contribute

Having real feedback on treatment adherence and effectiveness, Pharmora can enhance their market positioning, improve patient satisfaction.

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Utilize Distribution Channels

Divert Pharmora's Supply Chain partially to Patient-Centric State

Solution (1/4) - B: Direct-to-Customer (DTC) Model

1. Normal Operation Model
2. **Direct-to-customer (DTC) Model**

Digital

Onsite

Launching Product Model
Live Low. Live Strong

You're directed to Pharmora's page

Identification card

BUY

PRESCRIPTION

TIP-CHECK

You're having 60% sale – receive more via:

1. **Verify Identification:**
Secure customers' safety and Assure transparency

2. **Buy**
Where customers can see all the drugs' form, side effects & information

3. **Prescription**
Where customers can check for their prescription deliver's state

4. **Fact-check**
Where customers can check for their health state or fun fact related

5. **Contact Info**
Where customers can register to receive notifications of tips or promotions

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Launch Promotional Campaigns

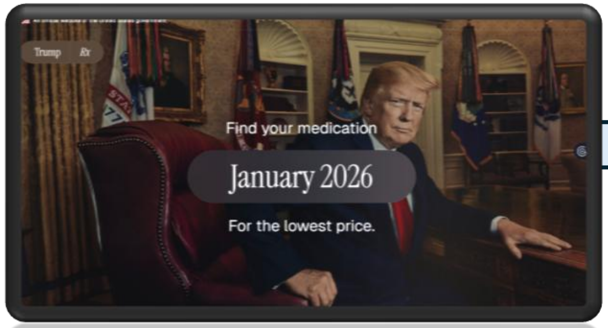
Pharmora's products should be launched cooperatively

Solution (1/4) – C : Launching Product to Market New Model

Normal Operation Model
Keep in touch with our target market

Digital
Onsite

Launching Product Model in 2026
Live Low. Live Strong



Low-priced

Recommended by GOV

Boost in Pharmora's Reputation

Disease-education
High Blood Pressure accounts for 50% of all heart disease; 40% of high Cholesterol people suffer from heart disease (WHF, 2025)

Dramatic Ad
"The more you know, the more likely you are to keep it pumping."

Campaign Website
Symptom checker
Personal stories humanizing struggle, creating trust

Engaged Patients
Community, Hashtags under campaign for patients to support others.

Experimental
Physical Installation: heart made of clocks

Pharmora

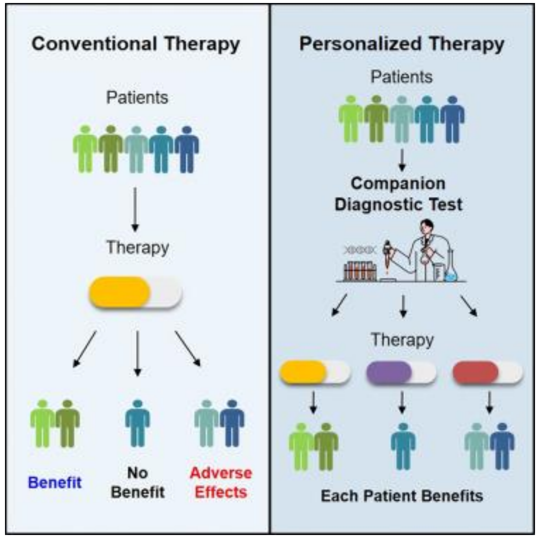
Serve Patients

Deploy "Therapy-as-a-Service" (TaaS) Model

Solution (2/4): Disease Management Ecosystems

1. Diagnosis

Companion Diagnostics

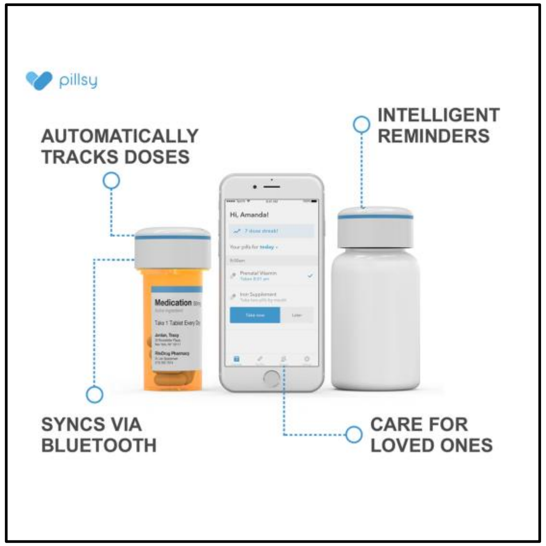


A medical test that provides information essential for the safe and effective use of a specific drug or biologic

Implications: Customers can see if they fit our medicine and personalized their conditions

2. Drug Purchase

Pharmora's Core Offering 2.0



Pillsy

- Bluetooth smart bottle
- Tracks days
- Sends reminders
- Notifies caregivers of missed doses

Implications: Help customers use Pharmora's drugs effectively

3. Telehealth

Direct-to-Consumer Telemedicine

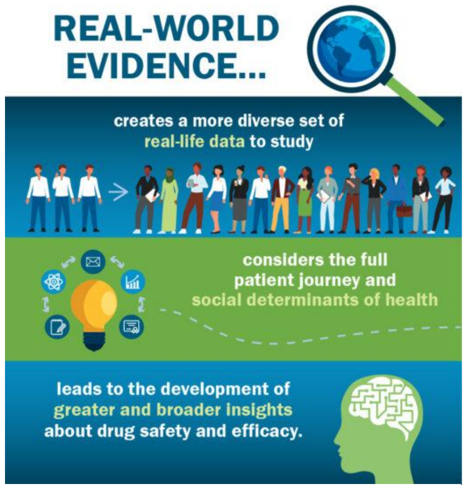
Store-and-Forward Telemedicine

Figure 1. Telehealth Platforms and Modalities.
Reproduced with permission from Mid-Atlantic Telehealth Resource Center

DTC = live, real-time virtual care
Store-and-Forward = recorded, reviewed later

Implications: Allow customers to actively interact with healthcare providers

4. Outcome Services



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Enforce

Artificial Intelligence (AI)

Mareana in Manufacturing Reduce COGS By Reducing Waste and SG&A by Lowering Working Hours, Digitalizing The Manufacturing Process

Solution (3/4): Integrate AI in Manufacturing + Impacts

Collab with Mareana's CPV to Pilot Run AI's effectiveness of 1-2 product lines for 6 -12 months

AI Data Integration | Anomaly detection | Problem Solving

- Digitize hand-written batch records
- Real-Time Insights for Monitoring
- Detects inconsistencies in batches, deviation management
- Flag uncertain entries for human review
- Data science studio for data analytics, reduce insight prep time

CPV

Improvement (vendor reported)

- 50% regulatory compliance
- 70% faster batch review
- 31% troubleshooting
- 30% waste reduction (Mckinsey & Company)

5% 📉 SG&A
10% 📉 COGS

Partnership with TriNetX in Drug Development Substantially Lower R&D and Provide Long-Term Edge Against Competitors, Transform Competency

Solution (4/4): Integrate AI in Drug Development + Impacts

20-100 → **Phase 1 clinical trial testing** → 3-12 months → 100-300 → **Phase 2 clinical trial testing** → 0.5-2 years → 300-3000 → **Phase 3 clinical trial testing** → 1-4 years → FDA → **Manufacturing**

32% of testing cost is recruitment BioPharma Dive	**$6500+** cost per recruit BioPharma Dive
40% dropout before completion BioPharma Dive	**$19,000+** cost per replace BioPharma Dive
80% fail to meet recruiting target Pubmed	**$8M** revenue lost per day due to delays Pubmed

TriNetX

Mass General Brigham

DEPARTMENT OF HEALTH & HUMAN SERVICES · USA

MAYO CLINIC

2013, Federated network (RWD & RWE)

- Data driven site identification + proactive recruitment
- 120+ health orgs (TriNetX,2025)

52% conversion rate increase for Crohn's Disease

37% conversion rate increase for Ulcerative Colitis

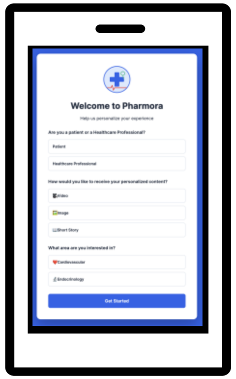
50 *biopharmaDive* new sites across 3 countries with 27 tier 1s in 1 day

→ **85%** conversion rate

$40 million in recruitment savings to recruit 100 patients

Integrate AI in Digital Marketing

Solution 1 Implementation: Generative AI to summarize and generate engaging content



Integrating AI in marketing creates a personalized experience for customers based on their preferences

Subscribe | Receive Daily News | Learn Anytime, Anywhere

Behind the Scenes

Input research
1. Clinical papers
2. Market research
3. Safety updates

Summarizes & Extracts Insights
1. Key findings trends
2. Medical insights
3. Regulatory-safe language

Script generation
Clear, storyboard-ready video script

Auto-generate scenes
Creates scenes, characters, and visuals

Export final video
Engaging animated explainer video ready for review or publishing

SPARK NLP

ChatGPT

VYOND

Financials

Impact

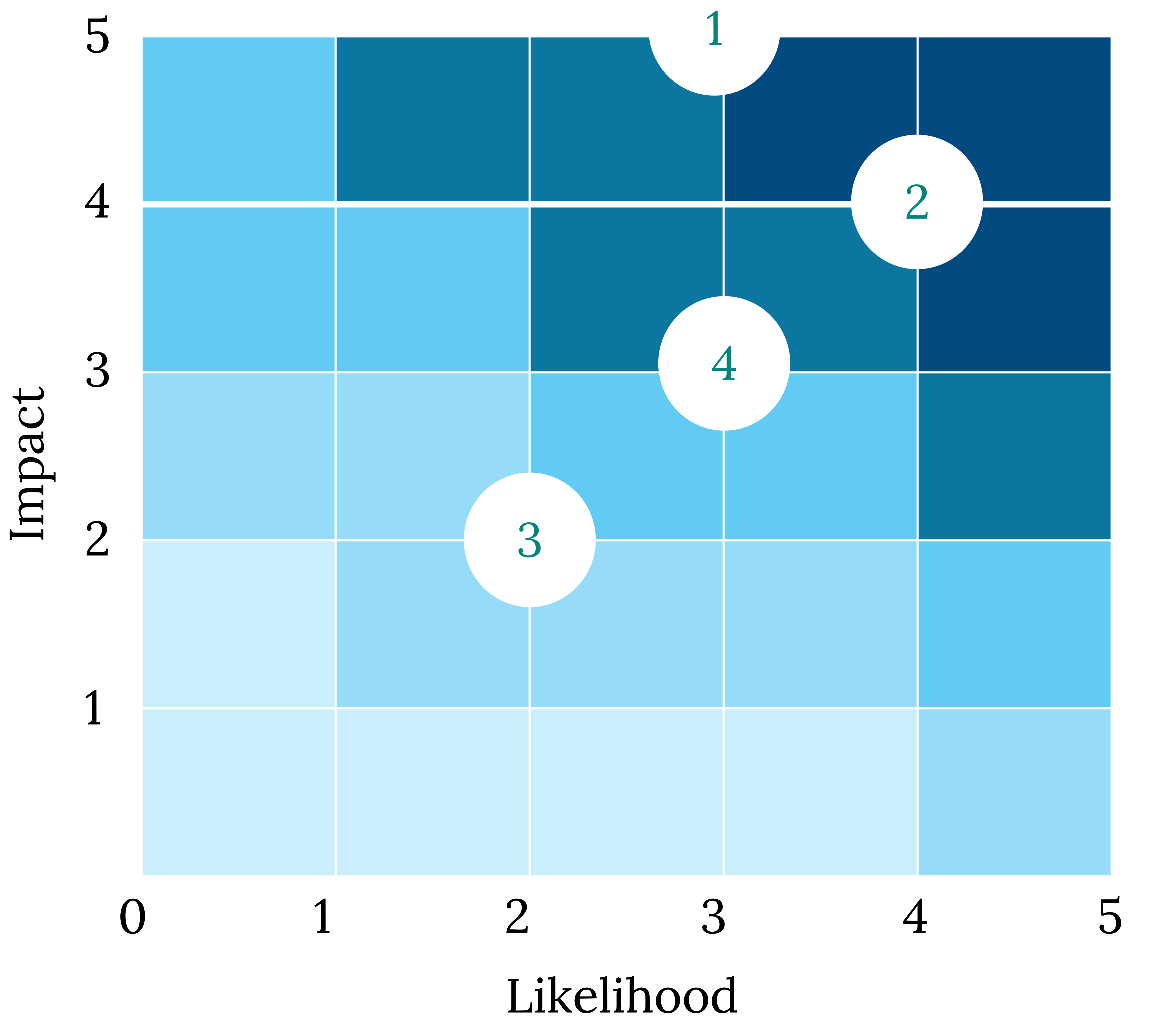
In short- to medium-term, technological integration and commercial expansion would bring significant customer base to Pharmora's products

Metric	Overall 5-Year % Change	Direction of Impact
Cardiovascular Revenue	**+40% to +60%**	Major growth from 3 drug launches + stronger digital/DTC [1] [2] [3]
Endocrinology Revenue	**+25% to +40%**	Driven by 2 late-period launches + education/DTC [1] [2] [3]
COGS	**−8% to −15%**	AI manufacturing efficiency + predictive automation [1] [6]
SG&A	**+5% to +12%**	Higher service + DTC investment offset by late efficiency [1]
R&D Expense	**−20% to −30%**	AI-enabled clinical trial acceleration & productivity [4] [5] [6]
Marketing Expense	**+10% to +20%**	Digital + DTC expansion (higher spend but higher ROI) [1] [2] [3]

[1] ZipDo 2025, [2] Gitnux, [3] DoctorMarketingMD, [4] Tecknoworks 2025, [5] BusinessToday, [6] Wifi Talents

Several Regulatory and Technological risks could hinder P.U.L.S.E.

Risks & Mitigation



	Risk	Mitigation
1	Regulatory and Compliance	Mareana + TriNetX follows SSoT, audit trails, have an AI compliance committee to ensure legality
2	Patient Data Privacy (HIPAA)	Business Associate Agreement with all AI vendors to enforce safety of patient's data
3	Damaging reputation by AI content + DTC	Involve professionals, highlight clinical evidence & patient exp, retain connection w/distributors
4	Over reliance on tech partners	Scale from collabs to partners to gradually being able to vertically develop our AI

Pharmora should focus on integrating AI and partnership in short-term, creating an open ecosystem in long-term.

Timeline Implementation

	2026	2027	2028	2029	2030
Digital Marketing	Preparing database	Operational			
Deploy "Direct-to-Consumer" Model	TrumpRx Website	Operational			
Launch Products to Market	2 cardio drugs		Cardiac Weight Control	Blood Sugar Control	Gene Therapy
Deploy "Therapy-as-a-service" Model	Regulatory Compliance	Pillsy integration Companion Diagnostic		Telehealth Partnership	
Partnership with TriNetX	Partnership		Long-term Partnership		
Integrate AI into Manufacturing	Collaboration 1 year	Partnership		Partnership	

Regulatory compliance before DTC

Campaign on New Product Release

Risk

Evaluation

Integrate AI to Create an Ecosystem

Long-term Vision: From Multi-channel to Omni-channel Model

SPARK NLP

VYOND

ChatGPT

1

Digital Marketing

3

Telehealth

Teladoc HEALTH

4

ORACLE Advertising and CX

Companion Diagnostics

Companion Diagnostics

TrumpRX

Companion Diagnostics

Direct-to-Consumers

2

Pillsy

Commercializing ⟷ **Real-world Data/Real-world Evidence** ⟷ **Manufacturing**

Pumping and Circulating Pharmora's Drugs Onto Customers

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Question

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J - LAMB

Pharmora

Thank You For Your Time!

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Pharmora

Appendix

Pharmora's Opportunity Lies In Approaching New Target Markets

Market Analysis – Consumer Insight

Key Trends

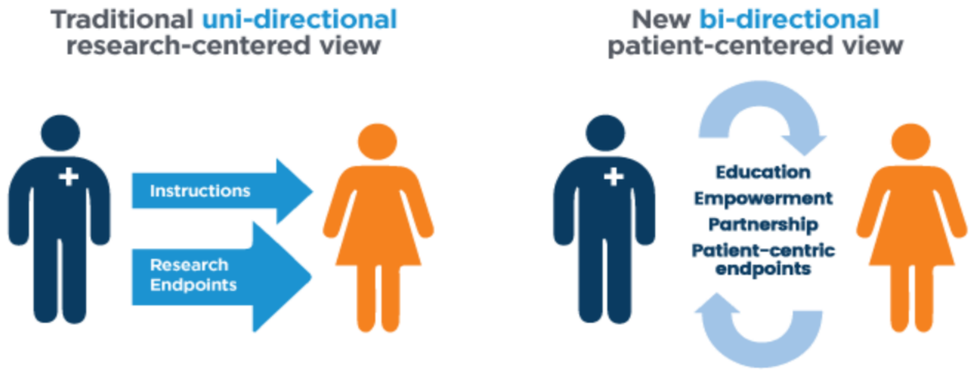
Therapy-as-a-service model gains tractions for personalized healthcare

"A recent Salesforce study found that 72% of consumers would be willing to receive communications and services from pharma companies that would 'make their lives easier,' yet only 21% reported being offered medication management tools or reminders." (Aquent, 2025)

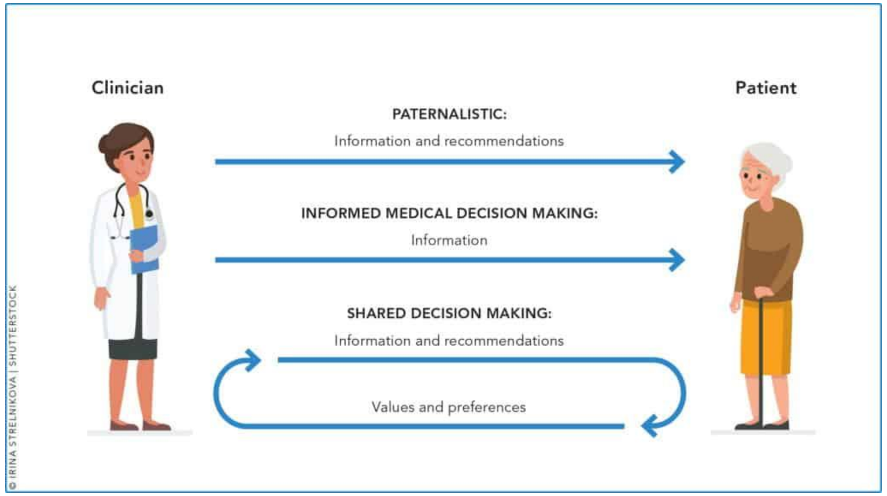
Direct-to-Consumer model likely benefits as uninsured patients increase

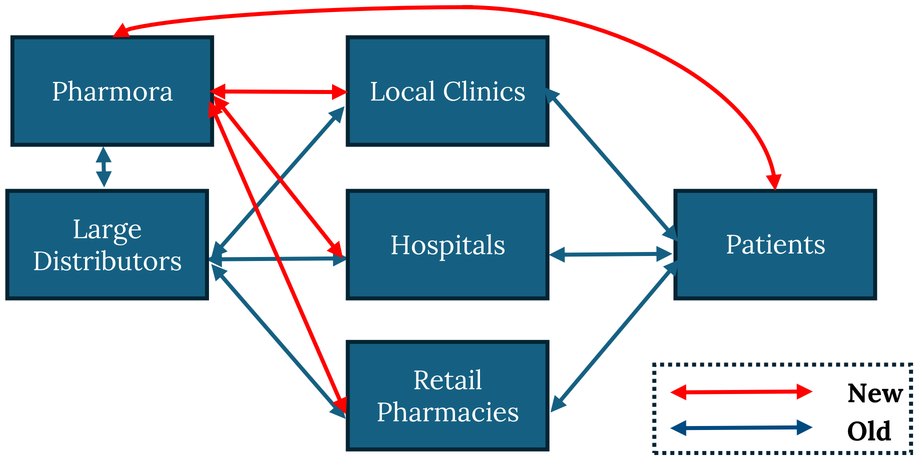
"An Additional 10M People Nationwide Could be Uninsured in 2034 Due To The Budget Reconciliation Package." (KFF, 2025)

Despite rising shared decision-making, HCPs still drive most patient drug decisions



Target Demographic





Utilize Cost Efficiency

Additional Solution Consideration: Cut SG&A and R&D cost

More applications of AI adoption would help save labor expenses on:

1. Sales Force Management
2. Marketing & Promotion
3. CRM & Call Centers
4. Finance & Accounting
5. Procurement
6. HR & Talent Management
7. Compliance & Legal
8. Medical Affairs

Overseas R&D Cost-sharing Agreement

Implications: Being able to reduce up to 40% SG&A expense in the next 5 to 7 years (The Hackett Group, 2024)

Implications: Being able to reduce R&D cost or increase R&D return due to a lower tax rate than the U.S. and generous tax credits

Partnership with TrinetX in Drug Development Substantially Lower R&D

Implementation

Partnership with TriNetX with milestone-based payments tied with output and royalties, ensuring minimal risks

2026

- Pilot cholesterol management phase 3 trial
- Pilot blood thinning phase 3 trial

- Establish baseline enrollment metrics
- Target: 25% faster site activation vs traditional methods
- Achieve 60% conversion rates
- 1000+ eligible patients

2027

- Scale recruitment for phase 3 trials
- Begin site feasibility for phase 2028 programs
- Mid-term partnership ROI review

- 70-80% conversion rates
- Exceed phase 3 enrollment by 15&

2028

- Manage 5 concurrent trials (two phase 3, three phase 2)

- 80-85% conversion rates across phase 2 trials
- 2500+ patients in total across active trials

2029

- Cholesterol management NDA submission

- Submit NDA 3-12 months ahead of projections
- $60 million in recruitment savings
- 15-18 months saved

2030

- Cholesterol management drug approval with TrinetX
- Partnership renewal
- Post market surveillance

- Multiple NDA submissions readily queued in 2031-2032

Enable allocation of R&D into future pipelines, aid in clinical insights to identify unmet needs and discover treatment gaps